US00PP31827P2

(12) **United States Plant Patent**
Ferguson et al.

(10) **Patent No.: US PP31,827 P2**
(45) **Date of Patent: Jun. 2, 2020**

(54) **STRAWBERRY PLANT VARIETY NAMED 'DRISSTRAWSEVENTYSIX'**

(50) Latin Name: ***Fragaria* x *ananassa***
Varietal Denomination: **DrisStrawSeventySix**

(71) Applicant: **Driscoll's, Inc.**, Watsonville, CA (US)

(72) Inventors: **Michael D. Ferguson**, Watsonville, CA (US); **Penny Nguyen**, Watsonville, CA (US)

(73) Assignee: **Driscoll's, Inc.**, Watsonville, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **16/501,758**

(22) Filed: **Jun. 4, 2019**

(51) **Int. Cl.**
A01H 5/08 (2018.01)
A01H 6/74 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./208**
CPC ***A01H 6/7409*** (2018.05)

(58) **Field of Classification Search**
USPC Plt./208
CPC A01H 6/7409; A01H 5/08
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP1,745	P	8/1958	Lang
PP3,981	P	11/1976	Bringhurst et al.
PP4,487	P	11/1979	Bringhurst et al.
PP4,538	P	5/1980	Bringhurst et al.
PP5,262	P	7/1984	Voth et al.
PP5,265	P	7/1984	Voth et al.
PP5,266	P	7/1984	Bringhurst et al.
PP5,300	P	10/1984	Johnson, Jr.
PP5,840	P	12/1986	Johnson, Jr. et al.
PP6,191	P	5/1988	Johnson, Jr. et al.
PP6,231	P	7/1988	Johnson, Jr. et al.
PP6,578	P	1/1989	Voth et al.
PP6,579	P	1/1989	Bringhurst et al.
PP7,024	P	9/1989	Johnson, Jr. et al.
PP7,172	P	2/1990	Voth et al.
PP7,522	P	5/1991	Johnson, Jr. et al.
PP7,614	P	8/1991	Bringhurst et al.
PP7,615	P	8/1991	Bringhurst et al.
PP8,086	P	1/1993	Nelson et al.
PP8,205	P	4/1993	Nelson et al.
PP8,649	P	3/1994	Sjulin et al.
PP8,661	P	3/1994	Bringhurst et al.
PP8,708	P	5/1994	Voth et al.
PP8,745	P	5/1994	Sjulin et al.
PP9,130	P	5/1995	Sjulin et al.
PP9,909	P	6/1997	Ackerman et al.
PP10,221	P	2/1998	Sjulin et al.
PP10,534	P	8/1998	Sjulin et al.
PP10,642	P	10/1998	Amorao et al.
PP11,035	P	8/1999	Mowrey et al.
PP11,277	P	3/2000	Gilford et al.
PP11,279	P	3/2000	Gilford et al.
PP11,522	P	9/2000	Amorao et al.
PP11,548	P	10/2000	Amorao et al.
PP11,554	P	10/2000	Sjulin et al.
PP11,639	P	11/2000	Mowrey et al.
PP12,186	P2	11/2001	Gilford et al.
PP12,436	P2	3/2002	Amorao et al.
PP12,577	P2	4/2002	Amorao et al.
PP12,817	P2	7/2002	Gilford et al.
PP12,899	P2	9/2002	Mowrey et al.
PP13,386	P2	12/2002	Mowrey et al.
PP13,469	P3	1/2003	Larson et al.
PP14,005	P3	7/2003	Amorao et al.
PP14,062	P3	8/2003	Amorao et al.
PP14,109	P3	8/2003	Gilford et al.
PP14,771	P3	5/2004	Amorao et al.
PP15,145	P2	9/2004	Mowrey et al.
PP15,308	P2	11/2004	Sjulin et al.
PP15,375	P2	11/2004	Mowrey et al.
PP15,435	P2	12/2004	Sjulin et al.
PP15,596	P2	3/2005	Amorao et al.
PP15,731	P2	4/2005	Amorao et al.
PP15,752	P2	5/2005	Gilford et al.
PP16,070	P2	10/2005	Gilford et al.
PP16,238	P2	2/2006	Amorao et al.
PP16,241	P2	2/2006	Mowrey et al.
PP16,298	P2	2/2006	Gilford et al.
PP16,299	P2	2/2006	Gilford et al.
PP16,475	P2	4/2006	Gilford et al.
PP16,558	P3	5/2006	López
PP18,000	P2	9/2007	Meulenbroek
PP18,040	P3	9/2007	Mowrey et al.
PP18,041	P3	9/2007	Gilford
PP18,458	P2	1/2008	Ferguson et al.
PP18,575	P3	3/2008	Amorao et al.
PP18,878	P2	6/2008	Mowrey et al.
PP19,240	P2	9/2008	Gilford et al.
PP19,673	P3	2/2009	Ferguson et al.
PP19,767	P2	2/2009	Shaw et al.
PP20,248	P3	9/2009	Rogers et al.
PP20,363	P2	9/2009	Chandler
PP20,701	P2	2/2010	Gilford et al.
PP20,731	P2	2/2010	Mowrey et al.
PP20,733	P2	2/2010	Mowrey et al.
PP20,735	P2	2/2010	Ferguson
PP20,775	P2	2/2010	Mowrey et al.
PP20,922	P2	4/2010	Gilford et al.
PP21,538	P2	11/2010	Gilford et al.
PP21,559	P2	12/2010	Ferguson et al.
PP21,762	P2	3/2011	Stewart et al.

(Continued)

OTHER PUBLICATIONS

Fear et al., Unpublished U.S. Appl. No. 15/998,014, filed Jun. 11, 2018, titled "Strawberry Plant Variety Named 'DrisStrawFiftyEight'".
Ferguson et al., Unpublished U.S. Appl. No. 15/998,317, filed Aug. 3, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixty'".
Ferguson et al., Unpublished U.S. Appl. No. 16/501,759, filed Jun. 4, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventySeven'".

(Continued)

Primary Examiner — Keith O. Robinson
(74) *Attorney, Agent, or Firm* — Morrison & Foerster LLP

(57) **ABSTRACT**

A new and distinct variety of strawberry plant named 'DrisStrawSeventySix', selected for its fruit flavor, fruit size, and yield, is disclosed.

6 Drawing Sheets

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP22,040	P3	7/2011	Stewart et al.
PP22,218	P2	11/2011	Ferguson
PP22,247	P2	11/2011	Ferguson
PP23,107	P2	10/2012	Ferguson et al.
PP23,148	P2	10/2012	Gilford et al.
PP23,377	P2	2/2013	Ferguson et al.
PP23,378	P2	2/2013	Pullen et al.
PP23,382	P2	2/2013	Ferguson et al.
PP23,383	P2	2/2013	Ferguson et al.
PP23,400	P2	2/2013	Ferguson et al.
PP23,401	P2	2/2013	Pullen et al.
PP23,459	P2	3/2013	Stewart et al.
PP23,506	P3	4/2013	Ferguson et al.
PP23,517	P3	4/2013	Ferguson et al.
PP24,096	P3	12/2013	Fear et al.
PP24,317	P3	3/2014	Ferguson et al.
PP24,333	P3	3/2014	Vitten et al.
PP24,395	P3	4/2014	Vitten et al.
PP24,533	P3	6/2014	Ferguson et al.
PP24,745	P2	8/2014	Vitten et al.
PP25,408	P3	4/2015	Vitten et al.
PP25,437	P3	4/2015	Vitten et al.
PP25,698	P3	7/2015	Ferguson et al.
PP25,699	P3	7/2015	Stewart et al.
PP25,747	P3	7/2015	Kibbe et al.
PP25,866	P3	9/2015	Ferguson et al.
PP26,800	P3	6/2016	Stewart et al.
PP26,801	P3	6/2016	Stewart et al.
PP26,802	P3	6/2016	Rodriguez Alcazar et al.
PP27,442	P2	12/2016	Kibbe et al.
PP27,645	P3	2/2017	Vitten et al.
PP27,682	P3	2/2017	Kibbe et al.
PP27,711	P3	2/2017	Vitten et al.
PP27,813	P3	3/2017	Ferguson et al.
PP29,289	P3	5/2018	Vitten et al.
PP29,728	P2	10/2018	Stewart et al.
PP29,729	P2	10/2018	Kibbe et al.
PP29,730	P2	10/2018	Kibbe et al.
PP29,731	P2	10/2018	Ferguson et al.
PP29,747	P2	10/2018	Vitten et al.
PP29,748	P2	10/2018	Vitten et al.
PP29,749	P2	10/2018	Stewart et al.
2003/0079263	P1	4/2003	Gilford et al.
2013/0276182	P1	10/2013	Fear et al.

OTHER PUBLICATIONS

Ferguson et al., Unpublished U.S. Appl. No. 16/501,428, filed Apr. 12, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventyFive'".

Jacobs et al., Unpublished U.S. Appl. No. 16/501,501, filed Apr. 18, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventy'".

Mendoza et al., Unpublished U.S. Appl. No. 16/350,139, filed Oct. 2, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtySeven'".

Mendoza et al., Unpublished U.S. Appl. No. 15/998,020, filed Jun. 12, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtyOne'".

Mendoza et al., Unpublished U.S. Appl. No. 15/998,169, filed Jul. 12, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtyThree'".

Mendoza et al., Unpublished U.S. Appl. No. 15/998,170, filed Jul. 12, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtyTwo'".

Mendoza et al., Unpublished U.S. Appl. No. 16/501,602, filed May 7, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSixtyEight'".

Pakozdi et al., Unpublished U.S. Appl. No. 16/501,198, filed Mar. 5, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventyThree'".

Pakozdi et al., Unpublished U.S. Appl. No. 15/998,015, filed Jun. 11, 2018, titled "Strawberry Plant Variety Named 'DrisStrawFiftyNine'".

Pakozdi et al., Unpublished U.S. Appl. No. 15/998,028, filed Jun. 14, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtyFive'".

Pakozdi et al., Unpublished U.S. Appl. No. 16/501,182, filed Mar. 5, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventyFour'".

Pakozdi et al., Unpublished U.S. Appl. No. 16/501,200, filed Mar. 5, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventyTwo'".

Stewart et al., Unpublished U.S. Appl. No. 16/350,432, filed Nov. 14, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtySix'".

Stewart et al., Unpublished U.S. Appl. No. 15/998,031, filed Jun. 14, 2018, titled "Strawberry Plant Variety Named 'DrisStrawSixtyFour'".

Stewart et al., Unpublished U.S. Appl. No. 16/501,449, filed Apr. 17, 2019, titled "Strawberry Plant Variety Named 'DrisStrawSeventyOne'".

STRAWBERRY PLANT VARIETY NAMED 'DRISSTRAWSEVENTYSIX'

Latin name: Botanical classification: *Fragaria* x *ananassa.*

Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawSeventySix'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawSeventySix'.

Strawberry plant variety 'DrisStrawSeventySix' originated from a cross between the proprietary female parent '108T181' (unpatented) and the male parent 'DrisStrawTwentySeven' (U.S. Plant Pat. No. 23,400). Progeny plants from this cross, including 'DrisStrawSeventySix', were asexually propagated via stolons in McArthur, Shasta County, Calif. in September of 2013. Strawberry plant variety 'DrisStrawSeventySix' was later specifically identified and selected in Ventura County, Calif. in October of 2013.

'DrisStrawSeventySix' was subsequently asexually propagated via stolons, and underwent further testing at a farm in Ventura County, Calif. for six years (2013 to 2019). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons.

'DrisStrawSeventySix' exhibits the following distinguishing characteristics when grown under normal horticultural practices in Ventura County, Calif.:

1. Spreading plant growth habit;
2. Serrate margin of terminal leaflet; and
3. Inflorescence on the same level as foliage.

'DrisStrawSeventySix' was selected for its fruit flavor, fruit size, and yield.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. The photographs are of plants that are 24 weeks old.

FIG. **1** illustrates whole fruit of variety 'DrisStrawSeventySix'.

FIG. **2** illustrates longitudinal sections of fruit of variety 'DrisStrawSeventySix'.

FIG. **3** shows the upper and lower surfaces of flowers of variety 'DrisStrawSeventySix'.

FIG. **4**A shows the upper surface of a leaf of variety 'DrisStrawSeventySix'. FIG. **4**B shows the lower surface of a leaf of variety 'DrisStrawSeventySix'.

FIG. **5** illustrates a plant of variety 'DrisStrawSeventySix'.

FIG. **6** illustrates plants of variety 'DrisStrawSeventySix'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawSeventySix'. The data which define these characteristics is based on observations taken in Ventura County, Calif. from 2013 to 2019. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawSeventySix' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawSeventySix' was taken from plants that were 24 weeks old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The RHS Colour Chart of The Royal Horticultural Society of London (RHS) (2007 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary,* 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:
- *Species.*—*Fragaria* x *ananassa.*
- *Common name.*—Strawberry.
- *Denomination.*—'DrisStrawSeventySix'.

Parentage:
- *Female parent.*—Proprietary variety '108T181' (unpatented).
- *Male parent.*—'DrisStrawTwentySeven' (U.S. Plant Pat. No. 23,400).

Plant:
- *Height.*—24.1 cm.
- *Diameter.*—38.9 cm.
- *Number of crowns per plant.*—5.
- *Growth habit.*—Spreading.

Stolon:
- *Average number of daughter plants per square foot.*—10.
- *Diameter at bract.*—3.3 mm.
- *Anthocyanin coloration.*—Weak.
- *Anthocyanin color.*—RHS 173A (Dark reddish orange).

Leaf:

Number of leaflets.—Three only.

Color of upper surface.—RHS NN137B (Greyish olive green).

Variegation.—Absent.

Terminal leaflets.—Length: 6.53 cm. Width: 5.58 cm. Length/width ratio: 1.2. Number of teeth/terminal leaflet: 22. Shape of base: Obtuse. Margin: Serrate. Shape in cross section: Concave.

Petiole:

Length.—16.9 cm.

Diameter.—3.65 mm.

Attitude of hairs.—Horizontal.

Bract frequency (number present on each petiole).—0.17.

Petiolule:

Length.—8.75 mm.

Diameter.—2.14 mm.

Stipule:

Length.—3.57 cm.

Width.—16.34 mm.

Anthocyanin coloration.—Weak.

Anthocyanin color.—RHS 36D (Pale yellowish pink).

Inflorescence:

Position in relation to foliage.—Same level.

Pedicel.—Attitude of hairs: Horizontal.

Flower.—Flower diameter (petal tip to petal tip on non-flattened flower): 16.69 mm. Arrangement of petals: Overlapping. Stamen: Present. Typical and observed number of flowers per plant: 68.

Petal.—Length: 11.51 mm. Width: 11.49 mm. Length/width ratio: 1.0. Typical and observed petal number: 5. Color of upper side: RHS NN155D (White).

Calyx.—Diameter (sepal tip to sepal tip, measured on back of flower): 34.30 mm.

Sepal.—Length (sepal tip to point of attachment to receptacle): 14.62 mm. Width: 5.30 mm. Typical and observed sepal number: 10.60.

Fruit:

Length.—53.7 mm.

Width.—45.1 mm.

Length/width ratio.—1.2.

Fruit hollow length.—28.9 mm.

Fruit hollow width.—11.1 mm.

Fruit hollow length/width ratio.—2.6.

Shape.—Conical.

Color.—RHS N45A (Moderate red).

Position of achenes.—Level with surface.

Position of calyx attachment.—Level with fruit.

Attitude of sepals.—Upwards.

Color of flesh (excluding core).—RHS 42B (Strong reddish orange).

Color of core.—RHS N34D (Moderately reddish orange).

Production:

Flowering interval.—January to May.

Harvest interval.—February to June.

Type of bearing.—Not remontant.

Productivity.—7,891 kg to 11,873 kg of fruit per acre per season from five-month-old plants when grown in Ventura County, Calif.

Resistance to diseases, pests, and abiotic stress: Two-spotted spider mite (*Tetranychus urticae*): Moderately resistant. Botrytis fruit rot (*Botrytis cinerea*): Moderately susceptible. Powdery mildew (*Podosphaera macularis*): Moderately resistant. Anthracnose crown rot (*Colletotrichum acutatum*): Moderately resistant.

COMPARISON WITH PARENTAL AND COMMERCIAL VARIETIES

'DrisStrawSeventySix' differs from the proprietary female parent '108T181' (unpatented) in that 'DrisStrawSeventySix' has higher yield, better flavor, larger berry size, and more vigorous canopy as compared to '108T181'.

'DrisStrawSeventySix' differs from the male parent 'DrisStrawTwentySeven' (U.S. Plant Pat. No. 23,400) in that 'DrisStrawSeventySix' has higher yield, better flavor, larger berry size, and more vigorous canopy as compared to 'DrisStrawTwentySeven'.

'DrisStrawSeventySix' differs from the commercial variety 'DrisStrawThirtySix' (U.S. Plant Pat.No. 25,698) in that 'DrisStrawSeventySix' has a spreading plant growth habit, a serrate margin of terminal leaflet, inflorescence on the same level as foliage, and achenes level with fruit surface, whereas 'DrisStrawThirtySix' has a semi-upright plant growth habit, a crenate margin of terminal leaflet, inflorescence above foliage, and achenes below fruit surface.

'DrisStrawSeventySix' differs from the commercial variety 'DrisStrawFortySeven' (U.S. Plant Pat. No. 27,813) in that 'DrisStrawSeventySix' has a spreading plant growth habit, an obtuse shape of base of terminal leaflet, inflorescence on the same level as foliage, and an upwards attitude of sepals on fruit, whereas 'DrisStrawFortySeven' has a semi-upright plant growth habit, a rounded shape of base of terminal leaflet, inflorescence above foliage, and an upwards to outwards attitude of sepals on fruit.

We claim:

1. A new and distinct variety of strawberry plant named 'DrisStrawSeventySix' as shown and described herein.

* * * * *

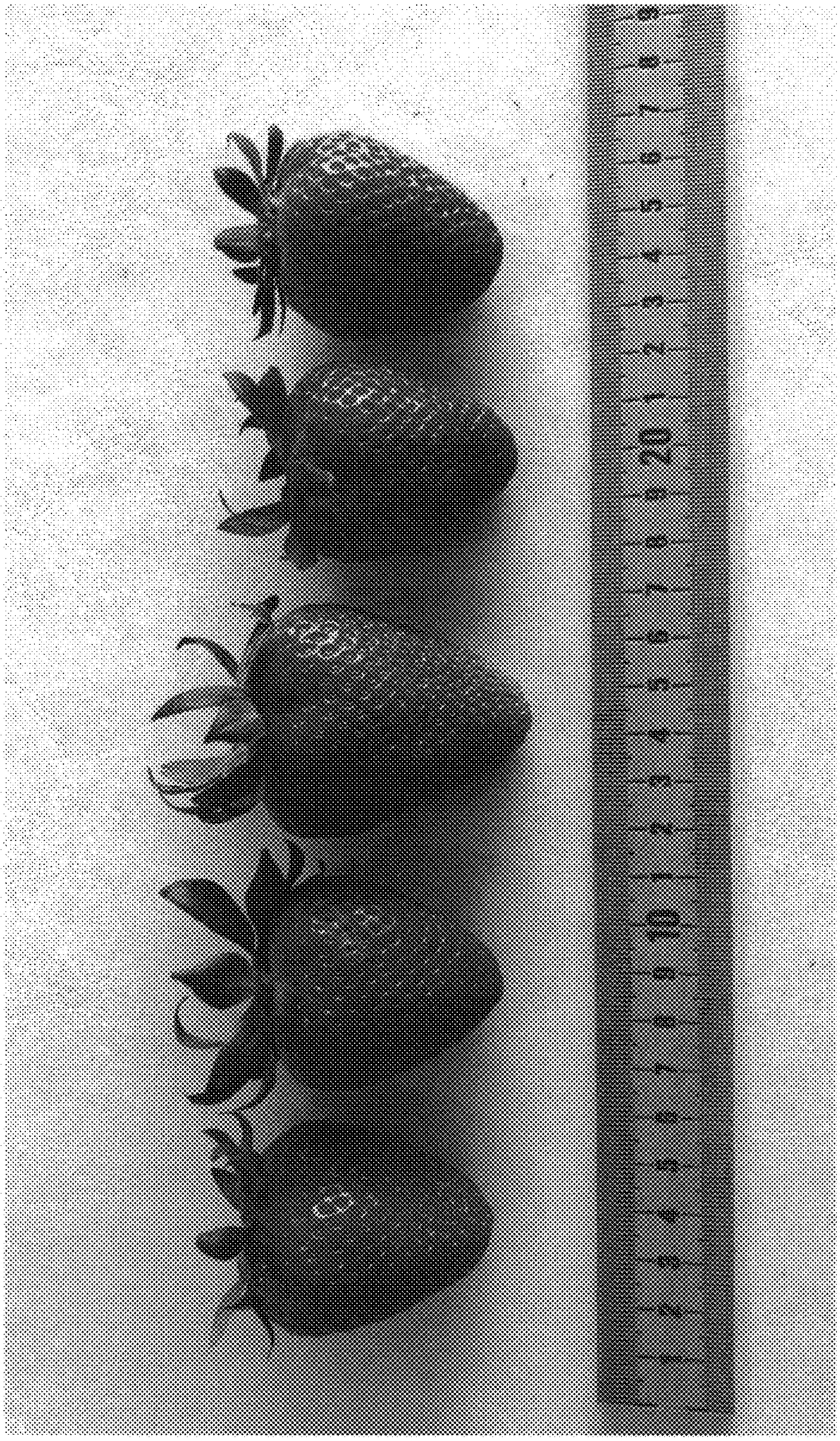

FIG. 1

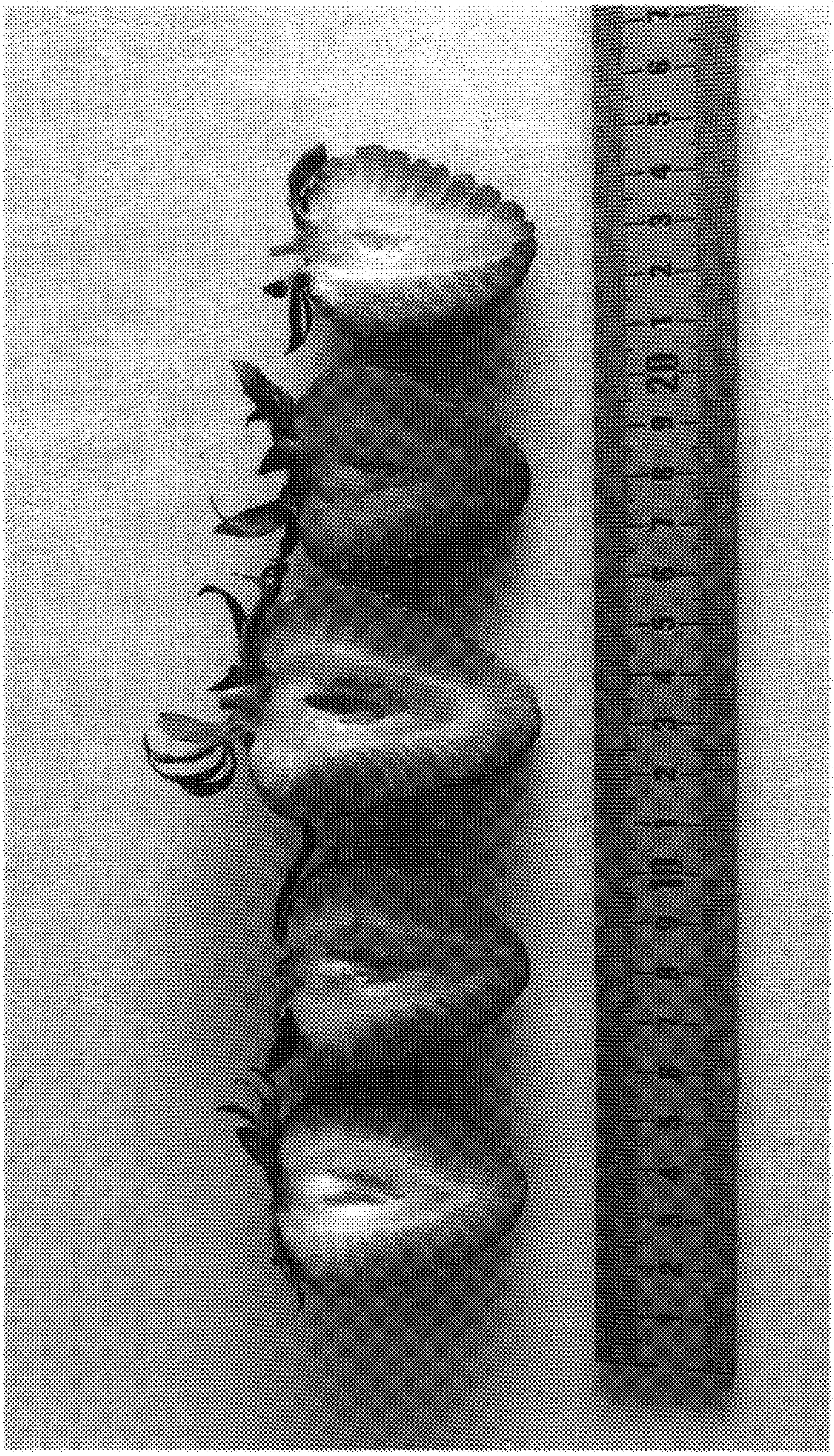

FIG. 2

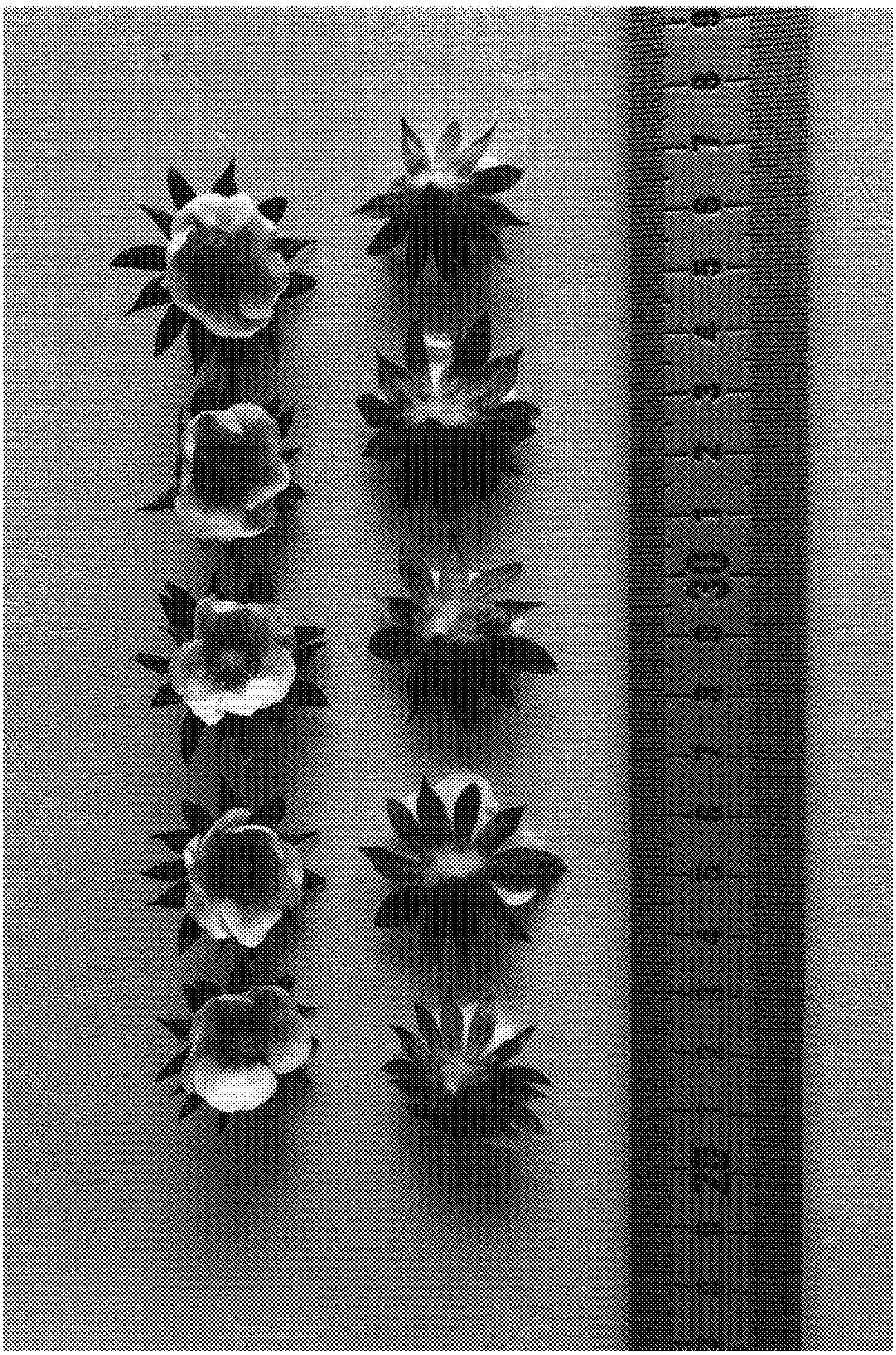

FIG. 3

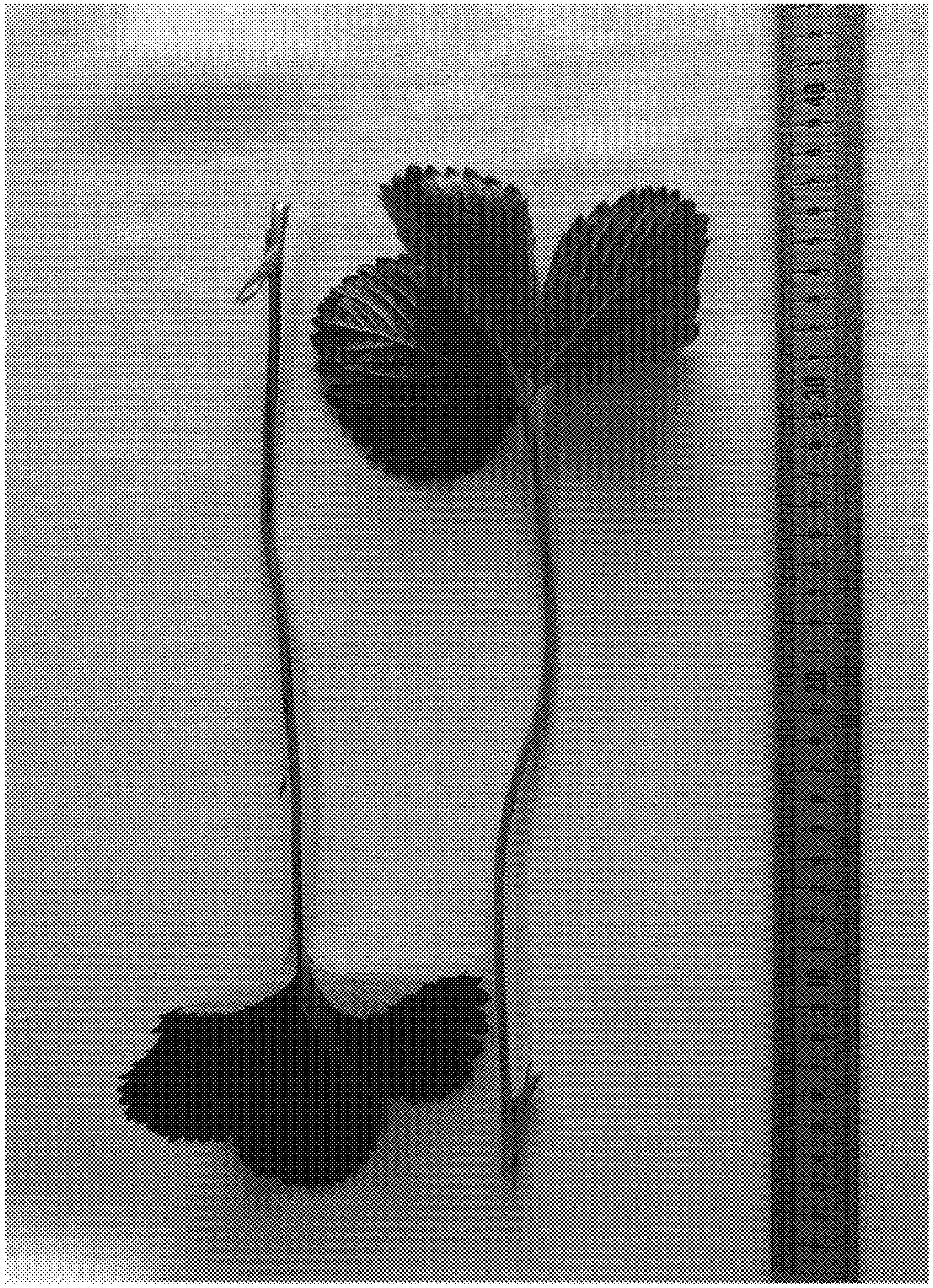

FIG. 4A

FIG. 4B

FIG. 5

FIG. 6

UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : PP31,827 P2
APPLICATION NO. : 16/501758
DATED : June 2, 2020
INVENTOR(S) : Michael D. Ferguson et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

On the Title Page

Item "(72) Inventors", left-hand column, Lines 9-11:
Delete "Michael D. Ferguson, Watsonville, CA (US); Penny Nguyen, Watsonville, CA (US)"
Insert -- Michael D. Ferguson, Watsonville, CA (US); Phuong Nguyen, Watsonville, CA (US) --

Signed and Sealed this
Twenty-seventh Day of April, 2021

Drew Hirshfeld
Performing the Functions and Duties of the
Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office